

DmV SOFTWARE

Marine Data Lab

Scientific and Operational User Manual

TEOS-10 / GSW oceanographic analysis, on a Copernicus-aligned QC foundation

Complete edition, verified against application source code
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Prepared for the Hellenic Centre for Marine Research (HCMR)

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1. Purpose, Scope and Key Advantages

Marine Data Lab is an oceanographic measurement analysis and quality-control (QC) application for the processing, calculation, validation and review of marine environmental observations. The application accepts georeferenced observations together with sampling date/time, platform, survey information, monitoring profile, sampling-depth category and sixteen measured oceanographic parameters. It calculates derived seawater properties using the TEOS-10 Gibbs SeaWater (GSW) methodology through the application's native GSW-C interface, performs parameter-specific QC tests aligned with Copernicus Marine Service and WOCE conventions, assigns automatic QC flags, supports manual QC review, maintains an analysis history and produces a formatted PDF analysis report with an embedded QR verification code.

1.1 Key advantages

- **Fully autonomous, offline operation.** Marine Data Lab requires no internet connection to run. TEOS-10/GSW calculations, all QC tests, the coastline/land-mask index, the History Log, Settings coefficients and PDF report generation are all performed locally, from data and libraries bundled with, or stored alongside, the application itself. No cloud service, external API or network call is required for any core analytical function.
- **Field-deployable.** Because it has no network dependency, the application can be used directly aboard a research vessel, at a remote coastal station, or in any other location without reliable connectivity -- observations can be entered, calculated, quality-controlled and saved to the local History Log immediately at the point of sampling, with reports generated and printed or queued for sending once connectivity becomes available.
- **Dual-mode quality control.** Mode 1 validates a single observation on its own physical, spatial, regional and thermodynamic merits; Mode 2 additionally checks a new observation against the researcher's own accumulated historical record. The two modes are complementary rather than alternative (§3, §10, §11).
- **Standards-aligned.** QC flag semantics follow the Copernicus Marine Service / WOCE 0-9 vocabulary; seawater thermodynamics follow TEOS-10/GSW; QC results are fully traceable per parameter and per test (§3, §12, §18.6).
- **Complete audit trail.** Every automatic QC decision is decomposable into its individual test results, each carrying its own applicability, execution status and QC flag, together with the automatic, manual and final QC values -- so a reviewer can see exactly how any flag was reached.
- **Bilingual, theme-adaptive, resolution-independent interface.** English and Greek localisation, light and dark themes, and an interface that scales continuously with window height rather than targeting one fixed screen resolution (§4).
- **Self-contained scientific reporting.** Each analysis can be exported as a formatted PDF report carrying a SHA-256 verification hash and a QR code, generated natively without any external report-writing service (§15).

1.2 The two analytical modes

- **Mode 1 -- Single Observation QC:** analysis of one observation using physical, spatial, regional, operational-range and TEOS-10 consistency tests.
- **Mode 2 -- Contextual / Historical QC:** analysis of one observation in relation to previously stored observations selected according to temporal, spatial and vertical-profile criteria.

Marine Data Lab is implemented as a JavaFX desktop application (Java 17).

1.3 Outlook: towards automatic parameter input

Measured-parameter entry is currently manual, by design, so that the application places no assumption on the instrument, sensor or logging format used by the researcher. As part of future adaptation and modernisation of the application, manual entry of the measured parameters may be supplemented or replaced by automatic input, tailored to the specific data carrier and file/data format produced by the measuring instrument in use (for example, direct import from a CTD, sensor logger or instrument export file). This would extend, rather than change, the QC and calculation architecture described in this manual: every test and formula described below would continue to operate identically on the resulting parameter values, whether they were typed in manually or imported automatically.

2. Application Architecture and Scientific Libraries

The principal application consists of the following components:

2.1 GSW-C / TEOS-10 native interface

Reached through a Java class `com.ocean.gsw.Gsw` and a JNI bridge to a compiled GSW-C library -- the official C implementation of the TEOS-10 Gibbs SeaWater oceanographic toolbox. All thermodynamic seawater calculations (Absolute Salinity, Conservative Temperature, potential temperature, density, potential density, sound speed, expected conductivity, oxygen solubility, depth-from-pressure) are delegated to this library rather than reimplemented in Java, ensuring the results are numerically identical to the reference TEOS-10 implementation used across the international oceanographic community, including by Copernicus Marine Service in-situ TAC and reanalysis products.

2.2 Land-mask and coastline service

`LandMaskService` uses **Gson** to parse a bundled GeoJSON coastline resource (`/geo/custom.geo.json`) and **JTS (Java Topology Suite)** -- specifically its Geometry model, STRtree spatial index and GeoJSON reader -- to build an in-memory spatial index of land polygons. This gives the application genuine point-in-polygon land/sea determination and true nearest-coastline distance calculation (§10.5), rather than a coarse bounding-box approximation, and does so entirely offline once the GeoJSON resource is loaded.

2.3 Report and imaging libraries

- **Apache PDFBox** -- native PDF construction (formatted text, tables, QC results, calculated results, notes, signature block, QR image) and printing, via PDFBox's pageable representation passed to the Java printing subsystem;
- **ZXing** -- generation of the 160×160-pixel QR verification code embedded in each report (§15.3);
- **Arimo** fonts -- bundled and loaded at start-up for consistent, licence-clear on-screen and in-report typography across platforms.

2.4 JavaFX interface layer

Controls, dialogs, the history table, theming and dynamic scaling are built directly on JavaFX, with no external UI framework dependency. Persistent application settings, the coefficient file and the history journal are stored as local text files in the user's application-data directory (§4.1), so the complete application state, including its full analytical history, resides on the local filesystem.

3. Standards Compliance Overview -- Copernicus, WOCE and TEOS-10

Marine Data Lab combines three distinct methodological layers, and understanding which layer a given number or test belongs to is essential to interpreting a QC result correctly:

Layer	Standard / origin	Where used
Seawater thermodynamics	TEOS-10 / GSW (IOC, SCOR, IAPSO, 2010; McDougall & Barker, 2011), called via the native GSW-C library	§8 -- Absolute Salinity, Conservative Temperature, density, sound speed, oxygen solubility
QC flag vocabulary	Copernicus Marine Service in-situ TAC QC flag scale, aligned with the WOCE 0-9 convention	§12 -- every automatic and manual QC flag produced by the application
QC test source classification	Application-defined, but explicitly categorised per test as Copernicus Core, Profile Structural, Marine Data Lab Scientific or Calculation Status	§10.9, §12.1 -- every individual QCResult
Operational and regional thresholds	Application/institution-defined, informed by regional oceanographic literature; editable by the user for the operational layer	§10.3, §10.4, Appendix A
Contextual/historical QC (Mode 2)	Application-defined statistical tests (spike, stuck value, rate of change, profile consistency, neighbour comparison) operating on the researcher's own History Log	§11, Appendix B

In short: the *meaning* of a QC flag (what “QC 3” or “QC 4” represents) is Copernicus/WOCE-standard and portable to any other Copernicus-compliant dataset; the physical seawater calculations that feed several of the tests are TEOS-10-standard and portable to any GSW-based workflow; but the specific numerical thresholds that decide which flag a given test assigns are Marine Data Lab's own scientific and operational configuration, and are documented threshold-by-threshold in §10, §11 and the Appendices below.

Where a specific test corresponds directly to a named Copernicus in-situ TAC QC test (for example, global range, regional range, spike or gross-range checks that mirror the Copernicus/Argo real-time QC test battery), this is noted explicitly in the relevant subsection of §10 and §11.

4. Initial Setup, Language, Theme and Scaling

4.1 Launch sequence

On start-up the application initialises, in sequence:

- the land-mask service;
- application directories;
- calculation-coefficient storage;
- locale and language resources;
- Arimo fonts;
- persistent application settings;
- the main JavaFX interface.

The application uses the user's application-data directory for persistent storage: the application directory, history file, settings file, PDF output directory, templates file and calculation-coefficient file. Directories are created automatically as required, and all of this storage is local -- no server-side account or network storage is involved.

4.2 Language

Two interface languages are provided: English and Greek. The language is an application-level property; changing it dynamically re-localises interface labels, buttons, field descriptions, QC descriptions, tooltip text and generated report text.

4.3 Light and dark themes

Light and dark themes are provided and persist with the other application settings. Theme selection affects the main interface, input fields, combo boxes, the history list, dialogs, QC displays, tooltips, Settings, the Mode 2 context controls and the preview interface.

4.4 Dynamic scaling

The interface is designed around the height of the active JavaFX stage. Font sizes, control heights, padding, spacing, dialog dimensions and table elements are calculated dynamically from the current window height rather than fixed to one display resolution. The same principle applies to the main controls, Settings, the Edit dialog, the Mode 2 custom-context dialog, Preview, QC tooltips and tables. The Preview window additionally calculates the available page area and scales the report page so the complete page remains within the available display area.

5. Observation Header, Coordinates, Platform and Survey

5.1 Header fields

The observation header contains: Latitude, Longitude, Sampling Date, Platform, Survey, Record ID and Researcher. The former internal field name *Research Vessel* is displayed to the user as *Platform*; the underlying application field is retained for compatibility with the existing data structure.

The sampling-date field uses the ISO-style UTC representation YYYY-MM-DDThh:mm:ssZ (for example 2026-08-19T12:30:00Z). The application assists entry of the separators and UTC suffix automatically, and performs date validation; a complete date/time value is required before the record can participate in Mode 2 temporal selection.

5.2 Geographic coordinates

Latitude and longitude are entered in decimal degrees. Latitude is restricted to -90° – $+90^{\circ}$ and longitude to -180° – $+180^{\circ}$, with input filters enforcing these bounds and permitting up to six decimal places. Coordinates feed several independent mechanisms: global coordinate validation, land/sea determination, distance-to-coast calculation, monitoring-profile consistency, TEOS-10 Absolute Salinity calculation, and Mode 2 spatial and profile selection.

5.3 Input auto-formatting and validation

Latitude, longitude, the sampling date/time and all sixteen measured-parameter fields are protected by dedicated input formatters (JavaFX TextFormatter filters) that constrain every keystroke to the correct format for that field as it is typed, rather than validating only after entry.

For the **sampling date**, the formatter automatically inserts the ISO separators and the trailing UTC Z as the digits are typed (→ YYYY-MM-DDThh:mm:ssZ), and any two-digit date/time component left incomplete when the field loses focus is automatically zero-filled. If, despite this, the sampling date is still not a valid, complete date/time when the researcher runs Analyse, the application does not proceed with the calculation: it returns focus to the sampling-date field with its content selected, so the researcher is taken straight back to correcting the date before anything else happens. Latitude and longitude are constrained to numeric input within their valid ranges (§5.2) as they are typed, and the sixteen measured-parameter fields are constrained to valid numeric input (with the parameter-appropriate sign and decimal handling) in the same way, so an invalid character or an out-of-format value is rejected by the field itself rather than discovered only later during QC.

5.4 Platform and Survey

Platform identifies the observational platform (research vessel, fixed station, buoy, autonomous platform, coastal station, or a laboratory/aquaculture platform) and is stored with the observation for inclusion in the report and verification data. **Survey** is a free-text field identifying the survey, campaign, cruise, monitoring programme or observational activity; no fixed scientific vocabulary is imposed.

5.5 Automatic, non-repeating Record ID

The Record ID field is read-only and generated automatically by the application; the researcher never types it. It has the form

YYYY-NNNN (for example, 2026-0143)

where YYYY is the current calendar year and NNNN is a four-digit, zero-padded sequential counter starting at 1. The counter is held in memory for the running session and is also written to, and reloaded from, the application's Settings file, so it persists across restarts of the application rather than resetting to 1 every time it is launched. Each time an observation is saved to the History Log, the application generates the native PDF report using the Record ID that was displayed at the moment of saving, **then increments the counter by one and immediately regenerates the displayed Record ID field** for the next observation -- guaranteeing that no two saved records ever share the same Record ID within the same year, without requiring the researcher to manage numbering manually. The year component is recalculated from the system clock each time the displayed ID is refreshed, so the sequence naturally starts again at 1 for a new calendar year rather than continuing indefinitely.

6. Monitoring Profile and Sampling Depth

6.1 Monitoring profile

Five monitoring profiles are provided, with internal identifiers: **CW** — Coastal Waters, **OS** — Open Sea, **HB** — Harbour, **AQ** — Aquaculture, **MPA** — Marine Protected Area. The selected profile determines the operational Min-Max range applied in Mode 1 and the spatial interpretation of the observation (§10.5).

6.2 Sampling-depth categories

Category	Internal code	Pressure/depth interval
Surface	SFC	up to 5 m
Shallow	SHL	5–20 m

Category	Internal code	Pressure/depth interval
Midwater	MID	20–100 m
Deep	DEP	100–500 m
Very Deep	VDP	> 500 m

Actual depth is derived from pressure through GSW (not a simple 1 dbar $\approx$ 1 m approximation), and the Sampling-Depth Context test then compares the calculated depth with the selected category (§10.7).

7. Measured Parameters and Missing-Data Handling

Marine Data Lab processes sixteen primary measured parameters, in a fixed internal sequence used consistently by the QC engine, the history parser and the Mode 2 history-selection mechanism: Water Temperature, Salinity, Conductivity, Pressure, Density, Sound Velocity, Dissolved Oxygen, Turbidity, pH, Chlorophyll-a, Nitrate, Nitrite, Phosphate, Silicate, Ammonium, Fluorescence.

A blank scientific input is treated as *not measured*, not as a measured numerical zero: internally a blank field becomes NaN. This distinction matters because zero is a legitimate scientific value for several parameters, whereas a blank field means the parameter was not observed and must not be interpreted as a measured zero -- the Missing Value QC test relies on this distinction to produce the missing-data flag (QC 9). The calculation engine explicitly checks whether a field is blank before any conversion.

8. TEOS-10 / GSW Calculation Engine -- Formulas and Constants

Marine Data Lab passes Practical Salinity (SP), in-situ temperature (t), sea pressure (p), longitude and latitude to the GSW interface -- these are direct calls to the published, internationally standard GSW-C library rather than independent Java implementations of the TEOS-10 equations, so the numerical results are identical to those any other GSW-based tool (including Copernicus Marine Service processing chains) would produce from the same inputs. The calculation sequence is:

```

SP, p, lon, lat → SA
SA, t, p → CT
SA, t, p → Potential Temperature
SA, CT, p → in-situ density
SA, CT → potential density anomaly ( $\sigma_\theta$ )
SP, t, p → expected conductivity
SA, CT, p → sound velocity
SA, CT, p, lon, lat → oxygen solubility
measured O2 + SA, CT, p, lon, lat → oxygen saturation %

```

8.1 Absolute Salinity (SA)

Calculated from Practical Salinity, sea pressure, longitude and latitude via the TEOS-10 salinity conversion. Practical Salinity is not treated as identical to Absolute Salinity; the calculated SA is displayed as a derived result and used in all subsequent TEOS-10 calculations. The TEOS-10 standard itself defines this conversion as

$$SA = S_R + \delta S_A$$

where S_R is Reference Salinity, obtained from Practical Salinity through the fixed global Reference-Composition ratio $S_R/S_P = S_{SO}/35 \approx 1.004715$ ($S_{SO} = 35.16504 \text{ g kg}^{-1}$, the Standard Ocean Reference Salinity defined by the TEOS-10 Reference-Composition Salinity Scale, 2008), and δS_A is the Absolute Salinity Anomaly, a small, geographically varying correction that the GSW library interpolates from a global reference atlas using longitude, latitude and pressure. This anomaly term is the reason Absolute Salinity calculation requires coordinates and pressure and not just Practical Salinity.

8.2 Conservative Temperature (CT) and Potential Temperature

CT is calculated from SA, in-situ temperature and sea pressure, and becomes the thermodynamic temperature variable used by subsequent density and sound-speed calculations. The TEOS-10 standard defines CT from the specific potential enthalpy h^0 referenced to the sea surface:

$$CT = h^0(SA, \theta) / c_p^0$$

where θ is potential temperature (referenced to $p = 0$ dbar) and c_p^0 is a fixed reference value of the specific heat capacity of seawater ($c_p^0 = 3991.86795711963 \text{ J kg}^{-1} \text{ K}^{-1}$ in the TEOS-10 standard). Potential Temperature is calculated independently from the measured thermodynamic state and displayed separately. The three temperature quantities therefore have distinct meanings: in-situ temperature is the measured value at the observation pressure; CT is the TEOS-10 conservative thermodynamic temperature; Potential Temperature is referenced to the sea surface but, unlike CT, is not exactly conservative under mixing.

8.3 In-situ density and potential density anomaly

$$\rho = \text{GSW density}(SA, CT, p) \quad [\text{kg m}^{-3}]$$

The GSW density function uses the TEOS-10 specific-volume formulation -- the 75-term polynomial approximation of Roquet et al. (2015), the standard computational form of the Gibbs function underlying the official GSW-C library. Potential density anomaly is calculated at the standard reference pressure corresponding to σ_0 :

$$\sigma_0 = \rho(SA, CT, p = 0) - 1000 \text{ kg m}^{-3}$$

8.4 Expected conductivity and Sound Velocity

Expected conductivity is independently calculated from Practical Salinity, in-situ temperature and pressure using the inverse of the Practical Salinity Scale 1978 (PSS-78) conductivity ratio relation, allowing measured conductivity to be checked against a thermodynamically consistent value during QC (§10.8). Sound velocity is calculated from SA, CT and p through the GSW sound-speed function -- based on the same TEOS-10 Gibbs function used for density, following the standard algorithm of Wong and Zhu (1995) as implemented in GSW -- and expressed in m s^{-1} .

8.5 Dissolved oxygen

Oxygen solubility is calculated from SA, CT, pressure, longitude and latitude. The GSW oxygen-solubility function implements the algorithm of Garcia and Gordon (1992), fitted to the oxygen-solubility data of Benson and Krause; the official TEOS-10 documentation notes this algorithm is included in the GSW toolbox as oceanographic best practice but is not itself an IOC-approved TEOS-10 Working Group 127 algorithm. The application additionally calculates oxygen saturation percentage and oxygen saturation deficit:

$$\begin{aligned} \text{oxygen saturation \%} &= 100 \times (\text{measured } O_2 / \text{oxygen solubility}) \\ \text{oxygen saturation deficit} &= \text{oxygen solubility} - \text{measured oxygen} \end{aligned}$$

8.6 Physical validity constant used by the thermodynamic pipeline test

The temperature/salinity/density thermodynamic pipeline check (§10.8) applies a fixed minimum physical validity criterion of **900 kg m⁻³** for the calculated in-situ density -- a sanity floor well

below any physically plausible seawater density, used only to catch gross calculation failure rather than to express a scientific density threshold (compare the much tighter Gross Range density bounds of 990–1070 kg m⁻³, §10.2).

9. Nutrient-Derived Quantities

Nutrient-derived values are calculated directly from measured concentrations and do not use GSW. These derived ratios (particularly DIN and the DIN:PO₄ and DIN:Si ratios) correspond to standard biogeochemical diagnostic quantities also reported in Copernicus Marine Service BGC in-situ and reanalysis products, though the application computes them independently from its own measured inputs rather than retrieving them from Copernicus:

$$\begin{aligned} \text{NO}_x &= \text{NO}_3 + \text{NO}_2 \\ \text{DIN} &= \text{NO}_x + \text{NH}_4 \quad (\text{equivalently } \text{DIN} = \text{NO}_3 + \text{NO}_2 + \text{NH}_4) \\ \text{DIN:PO}_4 &= \text{DIN} \div \text{PO}_4 \\ \text{DIN:Si} &= \text{DIN} \div \text{Si} \\ \text{NO}_3:\text{PO}_4 &= \text{NO}_3 \div \text{PO}_4 \end{aligned}$$

If a denominator is zero, the corresponding ratio is not calculated and is displayed as unavailable.

10. Mode 1 -- Single-Observation Quality Control

Mode 1 is the single-observation QC engine. For each parameter the application constructs a `ParameterQCResult` containing individual `QCTestResult` objects, each recording the parameter, test type, test source, applicability, execution status, QC flag and an explanatory message. The result of every individual QC operation is therefore preserved, not only the final parameter flag.

The principal Mode 1 test catalogue comprises:

- Missing Value
- Global / Gross Range
- Operational Min-Max Range
- Regional Range
- Spatial Bounding (Coordinate Validity)
- Position on Land
- Sampling-Depth Context
- Pressure/Depth Consistency
- GSW Temperature Pipeline
- GSW Salinity Consistency
- GSW Conductivity Consistency
- GSW Density Consistency
- GSW Sound-Speed Consistency
- Oxygen Solubility
- Oxygen Saturation
- Climatology (defined but not currently executed — see §18.2)

The five Mode 2 contextual tests are also preserved in the QC architecture but are marked not applicable while Mode 1 is active -- see §11 for how the two modes complement each other.

10.1 Missing Value Test

Executed for every parameter. A blank field → QC 9 (executed); a present value → QC 1. Source category: Calculation Status. This corresponds directly to the Copernicus/Argo real-time “missing value” QC test.

10.2 Global (Gross) Range Test

Fixed physical bounds, compiled into the application and unaffected by the Settings operational-range table. A value outside these bounds produces QC 4. This test is the direct functional counterpart of the Copernicus/Argo real-time *Global Range Test* (source category: Copernicus Core) -- an implausible-anywhere-in-the-ocean physical sanity check applied before any regional or operational refinement.

Parameter	Minimum	Maximum
Water Temperature	−2.5 °C	40 °C
Salinity	0	42
Conductivity	0	70
Pressure	0 dbar	11,000 dbar
Density	990	1070 kg m ^{−3}
Sound Velocity	1350	1600 m s ^{−1}
Dissolved Oxygen	0	600
pH	6.5	9.5
Chlorophyll-a	0	200
Nutrients (each)	0	300
Turbidity	0	2000
Fluorescence	0	1000

10.3 Operational Min-Max Test

Uses the editable Settings coefficient table: five profile-specific min/max pairs per parameter (Coastal Waters, Open Sea, Harbour, Aquaculture, Marine Protected Area), selected by the current monitoring profile. Result: QC 1 inside the configured range; QC 3 outside it; QC 2 if the configuration itself is invalid; not applicable if no usable range is configured. Source category: Marine Data Lab Scientific. This deliberately separates fixed physical plausibility (§10.2, Copernicus-aligned) from editable operational suitability specific to the institution's monitoring programme. The complete default matrix is given in **Appendix A**.

10.4 Regional Range Test

Independent of the editable Settings matrix. Source category: Marine Data Lab Scientific, conceptually parallel to the Copernicus/Argo *Regional Range Test*, but with application-defined region envelopes rather than official Copernicus regional-product boundaries. The application first classifies the coordinates into a broad marine region:

Region	Latitude	Longitude
Mediterranean Sea	30–46°N	−6–36°E
Black Sea	40–47°N	27–42°E
Red Sea	12–30°N	32–43°E
Baltic	53–66°N	10–30°E

For the Mediterranean Sea classification, the application uses:

Parameter	Minimum	Maximum
Temperature	10	32
Salinity	33	40
Dissolved Oxygen	150	350
Chlorophyll-a	0	15
Nitrate	0	12
Nitrite	0	1.5
Phosphate	0	1
Silicate	0	15
Ammonium	0	5

For every other case -- Black Sea, Red Sea, Baltic, or coordinates outside all four named envelopes -- the application applies one shared, broader set of default limits:

Parameter	Minimum	Maximum
Water Temperature	-2.0	38.0
Salinity	0.0	42.0
Dissolved Oxygen	0.0	500.0
Chlorophyll-a	0.0	100.0
Nitrate	0.0	50.0
Nitrite	0.0	5.0
Phosphate	0.0	5.0
Silicate	0.0	200.0
Ammonium	0.0	20.0

A value outside the limits applicable to its classified region produces QC 3.

10.5 Spatial Validity, Land/Sea and Monitoring-Profile Spatial Logic

The **Spatial Coordinate (Bounding) Test** verifies latitude and longitude are not NaN, not both zero, and within $\pm 90^\circ/\pm 180^\circ$ respectively; invalid or uninitialised coordinates → QC 4, valid coordinates → QC 1. Source category: Profile Structural.

Land/sea determination does not use a crude bounding box. LandMaskService loads /geo/custom.geo.json, converts each geometry to a JTS geometry and inserts it into an STRtree spatial index; a measurement point (longitude, latitude) is then tested by point-in-polygon query against the indexed land polygons. **Distance to coast** is the geometric distance to the nearest indexed land geometry, converted from coordinate degrees to kilometres:

$$\text{distance (km)} = \text{distance (degrees)} \times 111.32$$

The constant 111.32 km/degree is the standard mean length of one degree of latitude on the WGS-84/spherical Earth approximation. A point already on land is returned as 0 km from the coast.

Monitoring-profile spatial rules (application-defined, not a direct lookup of individual HCMR sites):

Profile	Requirement
Harbour	on land, or within 0.5 km of the coast
Coastal Waters	at sea, and no more than 15 km from the coast
Open Sea	at sea, and more than 15 km from the coast
Aquaculture	must not be on land
Marine Protected Area	must not be on land

10.6 Pressure-to-Depth Conversion and Pressure/Depth Consistency

Pressure is interpreted as sea pressure in dbar. Actual depth is obtained via the standard GSW function `depthFromPressure(pressure, latitude)`, which applies the TEOS-10 standard depth/pressure relation (a latitude-dependent correction on the hydrostatic approximation, accounting for the variation of gravity with latitude) -- no 1:1 dbar-to-metre approximation is used; if the GSW calculation fails, actual depth becomes unavailable rather than being replaced with an approximate value.

Pressure is additionally tested independently (source category: Profile Structural):

- missing pressure → QC 9;
- negative pressure → QC 4;
- depth cannot be calculated → QC 2;
- negative calculated depth → QC 4;
- otherwise → QC 1.

10.7 Sampling-Depth Context Test

Compares calculated actual depth against the selected sampling-depth category (§6.2). A mismatch → QC 3; a valid match → QC 1; unavailable depth → not executed/not applicable. Source category: Profile Structural.

10.8 GSW Scientific Consistency Tests

Additional scientific consistency tests, implemented by Marine Data Lab around TEOS-10-derived reference values (source category: Marine Data Lab Scientific, built directly on Copernicus-aligned TEOS-10 outputs): salinity, conductivity, density, sound-speed and temperature-pipeline consistency, plus oxygen-solubility and oxygen-saturation checks.

Salinity consistency — requires temperature, salinity, conductivity and pressure; if any is unavailable the test cannot be executed.

$$SP_{\text{calculated}} = \text{GSW salinity}(\text{conductivity}, \text{temperature}, \text{pressure})$$

$$\Delta SP = |SP_{\text{calculated}} - SP_{\text{measured}}|$$

Condition	QC
$\Delta SP \leq 0.05$	1
$\Delta SP \leq 0.20$	2
$\Delta SP \leq 1.00$	3
$\Delta SP > 1.00$	4

Conductivity consistency:

$$\Delta C = |C_{\text{expected}} - C_{\text{measured}}| \quad (\text{expected conductivity from SP, T, p})$$

Condition	QC
$\Delta C \leq 0.5$	1
$\Delta C \leq 2.0$	2
$\Delta C \leq 5.0$	3
$\Delta C > 5.0$	4

Density consistency:

$$\Delta \rho = |\rho_{\text{expected}} - \rho_{\text{measured}}|$$

Condition	QC
$\Delta\rho \leq 0.5 \text{ kg m}^{-3}$	1
$\Delta\rho \leq 2.0 \text{ kg m}^{-3}$	2
$\Delta\rho \leq 5.0 \text{ kg m}^{-3}$	3
$\Delta\rho > 5.0 \text{ kg m}^{-3}$	4

Sound-speed consistency:

$$\Delta c = |c_{\text{expected}} - c_{\text{measured}}|$$

Condition	QC
$\Delta c \leq 2 \text{ m s}^{-1}$	1
$\Delta c \leq 8 \text{ m s}^{-1}$	2
$\Delta c \leq 20 \text{ m s}^{-1}$	3
$\Delta c > 20 \text{ m s}^{-1}$	4

Temperature thermodynamic pipeline — verifies that the sequence SP → SA → CT → density produces physically valid results: availability of temperature, salinity and pressure; successful calculation of SA, CT and density; and density not falling below the application's minimum physical validity criterion of 900 kg m^{-3} (§8.6). A successful pipeline receives QC 1.

10.9 Oxygen Tests

Oxygen Solubility Test — ratio of measured to theoretical solubility:

$$R = O_{2\text{measured}} / O_{2\text{solubility}}$$

- $R < 0$ or $R > 2.0 \rightarrow$ QC 4;
- $R > 1.30 \rightarrow$ QC 3;
- otherwise $\rightarrow$ QC 1.

Oxygen Saturation Test — independently calculated saturation percentage:

- saturation $< 0\%$ or $> 200\% \rightarrow$ QC 4;
- saturation $< 50\%$ or $> 130\% \rightarrow$ QC 3;
- otherwise $\rightarrow$ QC 1.

10.10 Mode 1 Automatic QC Aggregation

Mode 1 uses all applicable and executed tests. Precedence (highest to lowest): QC 9 → QC 4 → QC 3 → QC 2 → QC 1. Non-applicable or non-executed tests do not participate. If no applicable test has executed, the automatic result is QC 0.

11. Mode 2 -- Contextual / Historical Quality Control

Mode 2 extends QC from single-observation validation to contextual analysis against historical observations, using the application's own History Log as a scientific reference dataset. It does not run the Mode 1 test suite in its place: the parameter QC engine explicitly switches to the Mode 2 test assembly when Mode 2 is active. Five contextual tests are executed:

- **C11-Spk — Temporal Spike**
- **C12-Stk — Stuck Value**
- **C13-ROC — Rate of Change**
- **C14-Prf — Profile Consistency**

- **C15-Nbr — Neighbour Comparison**

C11 and C13 correspond conceptually to the Copernicus/Argo real-time *Spike Test* and *Rate of Change Test* respectively, and C12 corresponds to the Copernicus/Argo *Stuck Value (Flat Line) Test*; all three are re-implemented here against the researcher's own historical record rather than a fixed shipboard time series, which is why they are classified as Marine Data Lab Scientific tests operating in a Mode-2-specific context rather than as Copernicus Core tests.

11.1 Mode 1 and Mode 2: how the two modes complement each other

Mode 1 and Mode 2 are not alternative ways of reaching the same conclusion -- they answer two different scientific questions about the same observation. Mode 1 asks: *"Is this observation physically and operationally plausible on its own terms?"* -- gross physical limits, TEOS-10 internal consistency, monitoring-profile suitability and spatial validity, none of which require any other observation to evaluate. Mode 2 asks a complementary question: *"Is this observation consistent with what has been measured before, at this place, this time of year, and this depth?"* -- a question Mode 1 cannot answer because it has no access to history. A value can pass every Mode 1 test (it is physically valid and within the operational range) and still fail Mode 2 (it represents an abrupt, unexplained departure from the local historical record), or conversely a value at the edge of a Mode 1 operational range may still pass Mode 2 cleanly if the local historical record already contains similar values. Running both modes on the same observation therefore gives a more complete picture than either alone: Mode 1 establishes absolute plausibility, Mode 2 establishes contextual plausibility.

Manual QC carries over between the two modes for direct comparison. The application keeps one set of per-parameter manual-QC values for the current session; these are only reset when the underlying measured value, sampling depth or monitoring profile is changed -- not when the analytical mode is switched. In practice, this means that if the researcher reviews an observation in Mode 1 and assigns manual QC values, then switches the same observation to Mode 2 without altering the measured fields, the Mode 2 QC review table opens with those same manual values pre-filled alongside the newly calculated Mode 2 automatic QC results -- making it straightforward to see, side by side, how the researcher's Mode 1 judgement compares with the independent Mode 2 contextual evidence, and to revise the manual QC accordingly.

11.2 Mode 2 Context

Automatic context (fixed): temporal window 30 days; spatial radius 10 km; spatial time window 7 days; depth tolerance 5 m. **Custom context** allows the researcher to adjust all four parameters directly, changing which historical observations are eligible and therefore potentially changing the Mode 2 result without changing the current measurement itself.

Only records identified as primary Mode 1 (M1) records in the History Log are ever used as the source population for Mode 2 selection -- records saved from a Mode 2 (M2) analysis are excluded from that pool. Since every Mode 2 analysis is itself performed on an observation that already has (or would otherwise duplicate) a corresponding Mode 1 entry, restricting selection to M1 records prevents the same physical observation from being counted twice in the historical evidence -- once directly and once again through its own Mode 2 record -- which would otherwise bias the temporal, spatial and vertical-profile statistics described in §11.3 and §11.4.

11.3 History Selection Mechanics

Temporal selection retains historical Mode 1 records within the configured temporal window (by absolute time difference in days) and sorts them chronologically. **Spatial selection** calculates the great-circle distance between current and historical coordinates using the haversine formula, with Earth radius $R = 6371.0088$ km (the IUGG mean Earth radius):

```

a = sin2(Δφ/2) + cos(φ1)·cos(φ2)·sin2(Δλ/2)
c = 2·atan2(√a, √(1-a))
distance = R × c

```

A historical record is retained only when distance ≤ configured spatial radius *and* time difference ≤ configured spatial time window. **Vertical-profile selection** retains records at the same geographic station area (within 0.5 km, within 24 hours) with a valid calculated depth and a valid value for the parameter, excluding the current record by Record ID; the selected records are ordered by actual depth.

Most Mode 2 tests require a minimum of **5 observations** (MODE2_AUTO_MIN_HISTORY_RECORDS) before execution. When insufficient history exists, the test is marked applicable but not executed, and the automatic Mode 2 QC for that parameter is then governed by the aggregation rules in §11.5, not treated as a passing test.

11.4 The Five Contextual Tests

C11 — Temporal Spike: median of same-parameter historical values within the temporal window; deviation of the current value from that median compared with the parameter's spike tolerance.

```
deviation = |current - median| → QC 4 if > tolerance, else QC 1
```

The median (not the mean) is used, making the reference less sensitive to isolated historical outliers.

C12 — Stuck Value: detects a flat-line sequence using two fixed observation windows (3 → suspect, 5 → bad). For a window:

```
range = max(window) - min(window); flat when range ≤ ε
```

Parameter-specific epsilon (ε): Water Temperature 0.05; Salinity 0.05; Pressure 0.5; all other parameters, 5% of that parameter's Mode 2 spike tolerance. A flat sequence of 5 observations → QC 4; of 3 observations → QC 3; otherwise QC 1.

C13 — Rate of Change: for each consecutive valid historical pair,

```
rate = |V2 - V1| / Δt (Δt in hours); maxRate = max(rate1, ..., raten)
```

maxRate compared with the parameter-specific rate tolerance: exceeding it → QC 3, otherwise QC 1. At least 5 historical observations are required.

C14 — Profile Consistency: requires a valid current calculated depth and at least 5 suitable profile observations. With V₁ = nearest shallower value, V₂ = current value, V₃ = nearest deeper value:

```
T = |V2 - (V1+V3)/2| - |(V3-V1)/2|
```

T compared with the parameter's profile tolerance: T > tolerance → QC 4, otherwise QC 1. Not executed if a bracketing observation exists on only one side of the current depth.

C15 — Neighbour Comparison: median of same-parameter values from historical observations within the spatial radius and temporal window (≥ 5 required):

```
difference = |current - neighbour median|
```

Exceeding the parameter-specific neighbour tolerance → QC 3, otherwise QC 1.

The complete per-parameter tolerance matrix (spike, stuck, rate, profile, neighbour) is given in Appendix B.

11.5 Mode 2 Automatic QC Aggregation

Considers only the five contextual tests. Precedence (highest to lowest): QC 9 → QC 4 → QC 3 → QC 2 → QC 0 → QC 1. If one or more applicable tests did not execute for lack of sufficient history, the result becomes QC 0 unless a higher-priority executed failure has

already been detected.

QC 0 in Mode 2 means that no complete contextual QC conclusion could be established from the available historical evidence — it does not mean the measurement itself is necessarily bad. This is fundamentally different from QC 4, which means an executed test identified bad data.

12. QC Flag Standard and Colours

The application displays the complete 0–9 QC vocabulary defined by the Copernicus Marine Service / WOCE standardisation:

Flag	Meaning
● 0	No QC performed
● 1	Good data
● 2	Probably good data
● 3	Bad data, potentially correctable
● 4	Bad data
● 5	Value changed
● 6	Not used
● 7	Nominal value
● 8	Interpolated value
● 9	Missing value

The circle colours above reproduce exactly the tooltip/legend colours used inside the application (JavaFX named colours GRAY, GREEN, GOLD, ORANGE, RED, DODGERBLUE, MEDIUMPURPLE, BROWN, TURQUOISE, DEEPPINK for flags 0–9 respectively). These colours are an application-interface convenience and are not part of the official Copernicus colour specification; the semantic meaning of a flag is determined by its numerical code, not by its display colour. The same legend, with the same colours, is available directly in the application as a hover tooltip (§13.1).

The current automatic QC engine directly produces flags 0, 1, 2, 3, 4 and 9. Flags 5–8 are part of the application's QC vocabulary and legend, available for manual assignment by the researcher, but are not automatically generated by the current Mode 1/Mode 2 aggregation algorithm.

12.1 QC Test Sources

Every individual QC result records its source category, distinguishing the conceptual origin of a test from its numerical result: **Copernicus Core** (tests that are direct functional counterparts of named Copernicus/Argo real-time QC tests -- Global Range, Missing Value, Climatology), **Profile Structural** (coordinate, pressure/depth and sampling-context validity), **Marine Data Lab Scientific** (TEOS-10 consistency checks, regional/operational ranges and the Mode 2 contextual tests), **Calculation Status** (whether a value or derived quantity could be computed at all).

12.2 Individual Test Result State

Every test result carries three logically independent states: **Applicability** (whether the test is scientifically applicable to the current parameter/context), **Execution** (whether the application had enough valid information to actually perform the test), and the resulting **QC flag**. This distinction matters especially in Mode 2: a temporal test may be applicable but not executed for lack of sufficient history, which must not be confused with a test that executed and passed.

12.3 Manual QC and Final QC

Each parameter carries a separate manual QC value that can override the automatically calculated flag. The internal model stores automatic QC, manual QC and final QC, where:

Final QC = Manual QC (if a manual value exists), otherwise Automatic QC

Manual QC values are stored in the history record and reconstructed for all sixteen parameters when a historical record is reopened, so the reviewed final state stays associated with the saved observation. As described in §11.1, manual QC values also persist across a Mode 1 ↔ Mode 2 switch within the same session, specifically to support side-by-side comparison of the two modes' evidence.

13. Interface Conveniences: Tooltips, Copy, Notes and Clear

13.1 Tooltips

Two dedicated hover tooltips are built into the interface. Hovering over a parameter's **QC flag label** opens the complete colour-coded 0–9 QC legend described in §12, so the meaning of any displayed flag is always one hover away. Hovering over the application's **title label** opens a second tooltip containing short descriptions of the depth/pressure relationship and the TEOS-10 temperature, salinity, conductivity, density and sound-speed calculations (§8), acting as an in-application glossary for the calculation engine. Both tooltips remain open indefinitely while hovered and follow the active light/dark theme.

13.2 Copy to clipboard

Any calculated result value displayed in the interface (Absolute Salinity, Conservative Temperature, density, sound velocity, oxygen and nutrient-derived quantities, etc.) can be copied to the system clipboard by clicking it; the application shows a brief confirmation popup and skips the action if the field currently shows no value. History Log entries can likewise be copied to the clipboard directly from the History list (single entry or current selection). When a PDF report is sent via *Send* (§15.2), the generated file itself is placed on the system clipboard as a file reference so it can be pasted directly as an attachment into an email client.

13.3 Notes

Each observation carries a free-text notes field, editable from the Edit dialog together with the QC review. Notes are saved together with the language they were written in, and are reloaded and displayed in that same language when the record is reopened, independent of the interface's current language setting -- so a note written in Greek stays in Greek even if the researcher has since switched the interface to English. Notes appear in the Preview and the generated PDF report alongside the observation and QC data.

13.4 Clear

The **Clear** control opens a confirmation dialog offering two independent actions, so that clearing the current form can never accidentally erase the History Log or vice versa:

- **Clear Data** — empties every field in the current observation form (coordinates, date, platform, survey, monitoring profile, sampling depth, all sixteen measured parameters, researcher) and resets all calculated results and QC displays to their empty state, without touching the History Log;

- **Clear History** — permanently erases the History Log file and the in-memory history list, without touching the current, unsaved form fields;
- **Cancel** — closes the dialog with no changes.

Because Clear History is irreversible and removes the entire reference population used by Mode 2 contextual QC, it should be used deliberately and not as a routine reset between observations -- Clear Data is the appropriate action for that.

14. History Log, Autofill and Search

14.1 History Log

The History Log is a persistent text file loaded into an observable JavaFX list on start-up. It is used for browsing previous analyses, searching, navigation, reopening observations, recovering manual QC values, and as the reference population for Mode 2 contextual analysis. The application retains up to **10,000** history records in memory; when the limit is exceeded, the oldest in-memory record is removed. A saved record contains the Record ID, current analysis date/time, Mode, coordinates, sampling date/time, survey, platform, monitoring profile, sampling depth, all sixteen measured values, calculated results, QC information, manual QC information, and notes.

14.2 History Search

The search field performs a plain, case-insensitive substring match against the visible text of each history-list entry. Matching rows are collected in order, the first match is selected and scrolled into view, and the up/down navigation buttons step between the collected matches -- a fast way to locate a previous observation by station name, platform, survey, date fragment or any other text visible in the entry, ahead of using Autofill (§14.3) or Mode 2 contextual analysis (§11) on it.

14.3 Autofill

Autofill is triggered by double-clicking, or pressing Enter on, a row the researcher has selected in the History list. Once triggered, the application:

- parses the selected history record's stored text fields;
- sets Researcher, Sampling Date, Latitude, Longitude, Platform and Survey from that record;
- sets the Monitoring Profile and Sampling Depth combo boxes from the record's stored codes;
- sets all sixteen measured-parameter fields (Water Temperature through Fluorescence) from the record's stored values;
- restores the record's saved notes and last-used language, then re-runs `calculate()` so the interface reflects the newly populated values;
- if the selected record was a Mode 2 record, switches the interface into Mode 2 and shows the Mode 2 context controls; otherwise it stays in Mode 1.

Automatic mode switching. This last step happens unconditionally and needs no separate action from the researcher: the analytical mode (Mode 1 or Mode 2) displayed and used after Autofill is always determined by which mode the selected History Log row (M1 or M2) was originally saved under, not by whichever mode the interface happened to be in immediately before Autofill was triggered.

Autofill copies every stored field from the single selected record, including the sixteen measured values, so it will replace any measurement values already present in the form. It is best used as a fast way to re-enter the header/context information of a known previous station (platform, survey,

monitoring profile, coordinates) or to reopen a past observation for review, and every populated field should be checked before running Analyse. Autofill does not perform any independent measurement and does not itself affect QC: once Analyse is run, QC is calculated solely from whatever values are present in the form at that moment, whether typed directly or introduced via Autofill.

14.4 Mode 2's use of the History Log

Separately from Autofill, Mode 2's five contextual tests draw their comparison population directly from the History Log using the temporal/spatial/vertical selection rules described in §11.3 -- this happens automatically whenever Mode 2 analysis runs, and does not depend on the researcher having used Autofill or Search first.

15. Manual QC, Reports, PDF, QR Code and Verification

15.1 Show Details / Edit Analysis

The Edit dialog presents observation information, measured parameters, calculated values, individual QC-test results, automatic/manual/final QC, notes, and application/save controls. The QC table is constructed dynamically for the active mode: in Mode 1 it shows the Mode 1 test columns; in Mode 2 it shows exactly the five contextual tests (C11–C15). Each parameter is a row.

15.2 Preview and PDF Report

Preview displays the report before it is finalised: title, Mode identification (M1/M2), observation header, QR verification code, measured parameters, calculated results, QC table, notes, researcher/signature section, and Apply/Save or Back-to-Edit controls. The window dynamically scales the report page to the available screen area. Mode 2 reports omit monitoring-profile/depth information from the header, since Mode 2 analysis is reported separately, while the measured-parameter section remains common to both modes.

The PDF itself is generated natively with Apache PDFBox (formatted text, tables, QC results, calculated results, notes, signatures, QR code and verification hash), and can be opened, printed or sent directly from the application.

15.2.1 Direct printing

Printing does not rely on opening the PDF in an external viewer first. The application loads the saved report file directly with PDFBox, wraps it in a PDFPageable object and hands that directly to the standard Java PrinterJob printing subsystem, which opens the operating system's native print dialogue for printer selection. The whole operation runs on a background thread so the interface stays responsive while the document is spooled to the printer.

15.2.2 Direct sending, with layered fallbacks

Sending copies the generated PDF file onto the system clipboard as a file reference (so it can be pasted as an attachment in any application) and then attempts to hand it directly to an e-mail client, trying each of the following in order and stopping at the first that succeeds -- without depending on Windows' default PDF viewer or any file-association setting at all:

- **Direct Outlook launch** -- the application checks a fixed list of known Microsoft Outlook installation paths (covering both 32- and 64-bit Office installs, Office “16” and “15” releases, under both Program Files and Program Files (x86)); for the first path found to exist, it launches Outlook directly with the `/a <file>` command-line switch, which opens a new message with the report already attached;
- **Operating-system mail facility** -- if no Outlook installation is found (or launching it fails), the application falls back to the Java Desktop API's mail action (`Desktop.getDesktop().mail()`), which hands off to whatever default mail handler the operating system itself provides;
- **Folder fallback** -- if neither of the above is available, the application opens the PDF's containing folder directly (using Windows Explorer's `/select` switch on Windows, or the platform's default file manager elsewhere) with the report file already highlighted, so the researcher can attach it manually -- and because the file was already placed on the clipboard at the start of this sequence, it can simply be pasted into the e-mail client without needing to browse for it again.

15.3 Verification Hash and QR Code

A verification hash is generated from key observation metadata -- Record ID, current date/time, latitude, longitude, sampling date, platform, survey and researcher, concatenated in that order -- using **SHA-256**; the displayed hash is the **first 16 hexadecimal characters, in uppercase**. The QR code, generated with ZXing's QRCodeWriter at **160 × 160 pixels**, carries the observation metadata and the verification hash, giving each report a compact, machine-readable identity.

16. Settings: Fixed versus Editable Thresholds

The application deliberately separates two categories of threshold, so that routine Settings changes cannot disable fundamental physical plausibility checks.

Fixed, compiled-in thresholds (not editable via Settings):

- gross physical ranges (§10.2);
- regional ranges (§10.4);
- Mode 2 minimum-history count, stuck-value observation counts, profile-station radius/time window;
- the Mode 2 parameter tolerance matrix and automatic-context defaults.

User-editable operational thresholds (Settings coefficient file):

The Min-Max operational-range table (§10.3), stored as ten numeric entries per parameter — min/max pairs for the five monitoring profiles — saveable to, and loadable from, the calculation-coefficient file. Changing a Settings range changes only the operational Min-Max test outcome; it does not change TEOS-10 calculations, GSW equations, gross physical limits, the Mode 2 tolerance matrix, or geographic calculations. For example, changing the Coastal Waters salinity range from 35.00–39.50 to another range may change the same historical measurement's Min-Max result from QC 1 to QC 3 or vice versa, while the fixed Gross Range result and all TEOS-10 values remain unchanged.

Likewise, changing the Mode 2 Custom context changes only which historical observations are accepted as contextual evidence; it does not change the current measured values, TEOS-10 calculations, gross physical ranges or Mode 1 operational ranges.

17. Recommended Scientific Workflow

- 1 Select language (English or Greek).
- 2 Select theme (light or dark).
- 3 Adjust display scale by resizing the window; interface elements scale with window height.
- 4 Optionally, use History Search to locate a related previous observation (same station, platform or survey).
- 5 Optionally, use Autofill on a selected History Log entry to copy its header/context information, or its complete field set, into the current form -- then review every populated field, including the measured parameters, before proceeding.
- 6 Select the analytical mode: Mode 1 for an individual observation, Mode 2 for contextual historical QC.
- 7 If Mode 2 is active, select Automatic or Custom context.
- 8 Enter or confirm observation metadata: latitude, longitude, sampling date/time, platform, survey, researcher, Record ID.
- 9 Select the monitoring profile.
- 10 Select the sampling-depth category.
- 11 Enter the available measured parameters. Leave genuinely unmeasured parameters blank — do not enter zero to indicate a parameter was not taken.
- 12 Run Analyse: the application validates the depth context, constructs the QC context, calculates actual depth, performs Mode 1 or Mode 2 QC, calculates TEOS-10-derived properties and updates the QC indicators.
- 13 Inspect the calculated results (Absolute Salinity, Conservative Temperature, density, sound velocity, oxygen and nutrient-derived quantities, and other outputs); click any value to copy it to the clipboard.
- 14 Inspect the overall parameter QC indicators; hover a QC flag for its legend.
- 15 Open Details/Edit to review each individual test result.
- 16 Apply manual QC where scientifically justified. If the observation was already reviewed in the other mode during this session, compare the carried-over manual QC against the current mode's automatic result (§11.1).
- 17 Add notes recording scientific interpretation, corrections or observations.
- 18 Preview the complete report.
- 19 Apply and Save to persist the analysis and its final QC state.
- 20 Generate, open, print or send the PDF report.

Interpreted end to end, Marine Data Lab is a layered QC system: Measurement → Physical plausibility → Operational suitability → Regional suitability → Geographic validity → Sampling-depth consistency → TEOS-10 thermodynamic consistency → (Mode 2:) Historical contextual consistency → Automatic QC → Scientific manual review → Final QC. The system does not replace scientific judgement; it separates automated evidence from the scientist's final interpretation. Automatic QC is the application's summary of the executed tests under the current configuration and available evidence; the detailed per-test review is the authoritative explanation of how that automatic flag was obtained; the final QC is the reviewed scientific classification after any manual intervention.

18. Scientific Interpretation and Methodological Notes

18.1 TEOS-10 scope and the oceanographic funnel

The GSW library implements the TEOS-10 Gibbs SeaWater framework, covering Practical and Absolute Salinity, Conservative Temperature, density, potential density, potential temperature, sound speed, dissolved gases and other thermodynamic properties, called through the Java interface rather than reimplemented. The GSW documentation also defines an “oceanographic funnel” within which its polynomial approximations are most accurate; the current Marine Data Lab Java source does not explicitly call a GSW funnel-validation function, so the application should not be described as independently performing an in-funnel test unless that functionality is added separately — the GSW library itself does provide it.

18.2 Climatology test — defined but not executed

The application contains a Climatology test object that is defined, applicable to selected parameters, but **not executed**, because no external parameter/latitude/longitude/depth climatology dataset is currently connected. It must not be interpreted as a real-time comparison against a Copernicus climatological product; the application deliberately reports that the reference dataset is not connected. The climatology hook is a placeholder for future development: as the application is further modernised, this test is intended, where scientifically appropriate, to be extended to cover as many of the measured parameters as possible and activated once the corresponding external climatology reference library or dataset has been connected, potentially as an additional contextual capability within Mode 2.

18.3 Parameters without independent TEOS-10 validation

TEOS-10 is fundamentally a thermodynamic seawater framework; it provides no independent physical validation algorithm for Turbidity, Nitrate, Nitrite, Phosphate, Silicate, Ammonium or Fluorescence. These parameters are evaluated through the application's range, regional, depth and contextual QC mechanisms rather than an invented TEOS-10 validation equation.

18.4 Limitations of the spatial dataset

The Java source specifies the land/coast algorithm but does not itself contain the coordinate vertices of the bundled GeoJSON resource; the precise coastline resolution, polygon source, generalisation and geodetic accuracy of the land mask depend on the actual GeoJSON file distributed with the application and cannot be established from the source code alone.

18.5 Manual QC persistence across modes -- interpretation caveat

Because manual QC values persist across a Mode 1 ↔ Mode 2 switch within the same session (§11.1, §12.3) but reset whenever a measured value, sampling depth or monitoring profile changes, a researcher comparing the two modes across separate sessions (rather than in one continuous review) will not see this carry-over -- each session starts from whatever manual QC was saved with the record, not from a fresh cross-mode default.

18.6 Final decision model and auditability

For each parameter the complete decision path is traceable: Input → (Mode 1: Missing Value → Gross Physical Range → Operational Min-Max → Regional Range → Spatial Validation → Land/Sea → Monitoring Profile → Sampling Depth → TEOS-10/GSW consistency → Oxygen tests where

applicable; or Mode 2: Missing Value → Historical selection → Temporal Spike → Stuck Value → Rate of Change → Vertical Profile Consistency → Spatial Neighbour Comparison) → Automatic QC → Manual QC if assigned → Final QC. The internal object model (ParameterQCResult containing a list of QCTestResult, each with parameter, test type, source, applicability, execution status, QC flag, message and optional spatial status) preserves a complete audit trail, and buildQCReviewRows() exposes this information as flattened parameter/test rows for detailed review.

This layered separation between measurement, calculation, automated evidence, contextual evidence and final scientific judgement is the central operating principle of Marine Data Lab.

Appendix A -- Complete Default Operational-Range Matrix

Default values of the editable Settings coefficient table (§10.3). All values are restored by the application if a parameter has no stored coefficient.

Water Temperature (°C)

Profile	Min	Max
Coastal Waters	10.0	30.0
Open Sea	12.0	28.0
Harbour	8.0	32.0
Aquaculture	10.0	30.0
Marine Protected Area	12.0	28.0

Salinity

Profile	Min	Max
Coastal Waters	35.0	39.5
Open Sea	37.5	39.2
Harbour	33.0	40.0
Aquaculture	35.0	39.5
Marine Protected Area	37.5	39.2

Conductivity

Profile	Min	Max
Coastal Waters	35.0	58.0
Open Sea	45.0	58.0
Harbour	30.0	60.0
Aquaculture	35.0	58.0
Marine Protected Area	45.0	58.0

Pressure (dbar)

Profile	Min	Max
Coastal Waters	0	200
Open Sea	0	4000
Harbour	0	35
Aquaculture	0	60

Profile	Min	Max
Marine Protected Area	0	1500

Chlorophyll-a

Profile	Min	Max
Coastal Waters	0.01	7.0
Open Sea	0.01	1.5
Harbour	0.1	25.0
Aquaculture	0.1	20.0
Marine Protected Area	0.01	2.0

Dissolved Oxygen

Profile	Min	Max
Coastal Waters	150	320
Open Sea	180	300
Harbour	80	350
Aquaculture	100	350
Marine Protected Area	180	300

pH

Profile	Min	Max
Coastal Waters	7.8	8.4
Open Sea	8.0	8.35
Harbour	7.4	8.5
Aquaculture	7.5	8.4
Marine Protected Area	8.0	8.35

Turbidity

Profile	Min	Max
Coastal Waters	0	15
Open Sea	0	2
Harbour	0	40
Aquaculture	0	25
Marine Protected Area	0	1.5

Nitrate

Profile	Min	Max
Coastal Waters	0	12
Open Sea	0	5
Harbour	0	40
Aquaculture	0	30
Marine Protected Area	0	3

Nitrite

Profile	Min	Max
Coastal Waters	0	1.5

Profile	Min	Max
Open Sea	0	0.5
Harbour	0	5
Aquaculture	0	4
Marine Protected Area	0	0.3

Phosphate

Profile	Min	Max
Coastal Waters	0	1
Open Sea	0	0.3
Harbour	0	5
Aquaculture	0	4
Marine Protected Area	0	0.2

Silicate

Profile	Min	Max
Coastal Waters	0	15
Open Sea	0	8
Harbour	0	30
Aquaculture	0	20
Marine Protected Area	0	6

Ammonium

Profile	Min	Max
Coastal Waters	0	3
Open Sea	0	0.5
Harbour	0	15
Aquaculture	0	10
Marine Protected Area	0	0.3

Fluorescence

Profile	Min	Max
Coastal Waters	0	10
Open Sea	0	3
Harbour	0	30
Aquaculture	0	25
Marine Protected Area	0	2.5

Appendix B -- Complete Mode 2 Parameter Tolerance Matrix

Exact values embedded in `createMode2ParameterMatrix()`. The “Stuck” column shown here is the fixed observation count (5); the stuck-value test's actual flat-line epsilon is described separately in §11.4 / C12.

Parameter	Spike	Stuck	Rate	Profile	Neighbour
Water Temperature	1.0	5	2.0	1.0	1.0
Salinity	0.5	5	0.5	0.5	0.5
Conductivity	0.8	5	1.0	0.8	0.8
Pressure	20.0	5	100.0	20.0	20.0
Density	0.5	5	1.0	0.5	0.5
Sound Velocity	5.0	5	10.0	5.0	5.0
Dissolved Oxygen	15.0	5	20.0	15.0	15.0
Turbidity	1.0	5	5.0	1.0	1.0
pH	0.15	5	0.10	0.15	0.15
Chlorophyll-a	0.5	5	1.0	0.5	0.5
Nitrate	2.0	5	5.0	2.0	2.0
Nitrite	0.5	5	1.0	0.5	0.5
Phosphate	0.5	5	1.0	0.5	0.5
Silicate	2.0	5	5.0	2.0	2.0
Ammonium	0.5	5	1.0	0.5	0.5
Fluorescence	0.5	5	2.0	0.5	0.5

Appendix C -- Numerical and Constants Reference Summary

Global physical limits (Gross Range test, §10.2)

Parameter	Minimum	Maximum
Water Temperature	−2.5 °C	40 °C
Salinity	0	42
Conductivity	0	70
Pressure	0 dbar	11,000 dbar
Density	990	1070 kg m ^{−3}
Sound Velocity	1350	1600 m s ^{−1}
Dissolved Oxygen	0	600
pH	6.5	9.5
Chlorophyll-a	0	200
Nutrients (each)	0	300
Turbidity	0	2000
Fluorescence	0	1000

Sampling-depth categories

Category	Range
Surface	≤ 5 m
Shallow	> 5–20 m
Midwater	> 20–100 m
Deep	> 100–500 m

Category	Range
Very Deep	> 500 m

Mode 2 fixed constants

Constant	Value
Minimum Mode 2 history	5 observations
Automatic temporal window	30 days
Automatic spatial radius	10 km
Automatic spatial time window	7 days
Automatic depth tolerance	5 m
Stuck-value suspect window	3 observations
Stuck-value bad window	5 observations
Profile station radius	0.5 km
Profile station time window	24 hours

TEOS-10 standard reference constants used in the calculation chain

Constant	Value / relation
Standard Ocean Reference Salinity, S_{0SO}	35.16504 g kg ⁻¹
Reference/Practical Salinity ratio, $S_{rSO}/35$	≈ 1.004715
Reference specific heat capacity, cp^0	3991.86795711963 J kg ⁻¹ K ⁻¹
Density specific-volume formulation	75-term polynomial (Roquet et al., 2015)
Sound-speed algorithm basis	Wong and Zhu (1995), as implemented in GSW
Oxygen-solubility algorithm basis	Garcia and Gordon (1992), fitted to Benson & Krause data

These are published TEOS-10 standard values implemented inside the linked GSW-C library; they are listed here for scientific completeness and are not application-specific constants (contrast with the application-defined thresholds elsewhere in this Appendix).

Application-specific geometric and validity constants

Constant	Value / relation	Used in
Degrees-to-kilometres conversion	111.32 km per degree	§10.5, coastline distance
Haversine Earth radius	6371.0088 km (IUGG mean)	§11.3, Mode 2 spatial distance
Minimum physical validity density	900 kg m ⁻³	§8.6, thermodynamic pipeline test
Verification hash algorithm	SHA-256, first 16 hex characters, uppercase	§15.3
QR code size	160 × 160 px	§15.3
Maximum in-memory history records	10,000	§14.1

Oxygen thresholds (Mode 1)

Test	Condition	QC
Oxygen Solubility	ratio < 0 or > 2.0	4

Test	Condition	QC
Oxygen Solubility	ratio > 1.30	3
Oxygen Solubility	otherwise	1
Oxygen Saturation	< 0% or > 200%	4
Oxygen Saturation	< 50% or > 130%	3
Oxygen Saturation	otherwise	1

End of manual.